

A COMPARATIVE APPROACH TO DEEPFAKE DETECTION

Fake Busters.

Five parallel detectors — spatial, frequency, anomaly, mesoscopic, and a hybrid SOA — stress-tested on 360,000 face crops to find which signal actually generalizes.

SOLUTION #1

Mohammed J. Darandari

Dual-Branch ConvNeXt V2 + Patch FFT

SOLUTION #2

Abdullah A. Alharbi

Autoencoder Anomaly Detection

SOLUTION #3

Mohammed A. Alqurayqiri

Hybrid Feature Fusion + RBF SVM

SOLUTION #4

Rakan F. Jalalah

MesoInception-4

SOA BENCHMARK

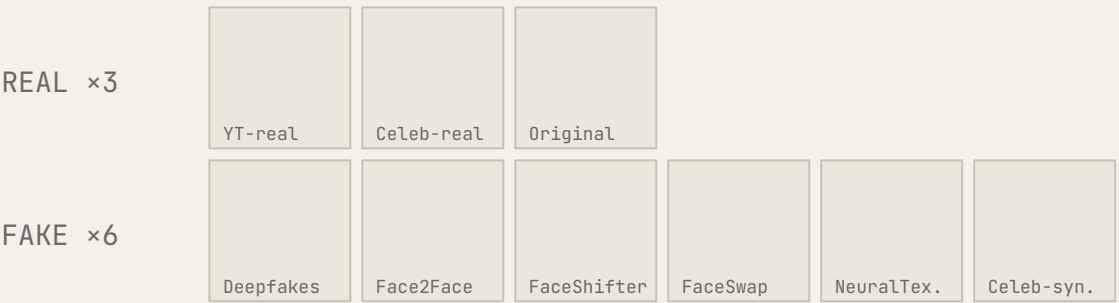
Muhannad A. Alzahrani

CLIP ViT-L/14 + GenD + D3 + PatchFFT

Detectors that ace one generator collapse on the next.

PROBLEM	Modern deepfake detectors latch onto generator-specific shortcuts. Accuracy looks great in-domain but collapses on unseen manipulations — the generalization gap.
WHY IT MATTERS	Synthetic media now drives identity fraud, election interference, and CEO impersonation scams measured in billions of dollars annually .
AUDIENCE	Newsrooms, KYC providers, social platforms, forensic analysts.
DATASET	FaceForensics++ U Celeb-DF v2 · 12K+ videos · ~360,000 RetinaFace-aligned 384×384 crops · 70/15/15 video-level split.

DATASET SNAPSHOT — 9 SOURCE CLASSES



Stratified by video ID — zero frame leakage between train / val / test.

Four diverse detectors and one SOA — to compare what each signal actually buys you.

GAP IN LITERATURE

Most published detectors report on **one generator family** with one architectural lens. Few studies hold the dataset, preprocessing, and split fixed and ask: *which signal — spatial, frequency, semantic, mesoscopic — pays off where?*

IMPLEMENTED PIPELINES

WHY A COMPARATIVE APPROACH

Five solutions on one shared 360K-image pipeline isolates the contribution of **each ML inductive bias** — not just whose model wins.

SOL #1 Dual-Branch ConvNeXt V2 + FFT DEEP CNN • SPATIAL + SPECTRAL Mohammed Darandari	SOL #2 Convolutional Autoencoder UNSUPERVISED • ANOMALY Abdullah Alharbi	SOL #3 Hybrid Feature Fusion + RBF SVM CLASSICAL ML • RESNET+DCT+LBP Mohammed Alqurayqiri	SOL #4 MesoInception-4 LIGHTWEIGHT CNN • MESOSCOPIC Rakan Jalalah	SOA CLIP ViT-L/14 + GenD + D3 + PatchFFT FOUNDATION • TRI-BRANCH HYBRID Muhannad Alzahrani
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ConvNeXt V2 + Patch-wise FFT.

DUAL-BRANCH CNN

ConvNeXt V2 Base + PatchFFT

Spatial branch (87.7M params, GRN blocks) sees pixel artifacts. Frequency branch FFTs a 12×12 grid of 32px patches (144 regions) and uses 4-head attention to find anomalous spectra. Fused 1280-dim → FC → 2-class.

WHY IT WORKS

Synthesis methods leave **periodic spectral artifacts** invisible in RGB. Spatial-only baseline caught 65.85%; adding the FFT branch lifted it to 77.22%.

SURPRISING FIND

The **FFT branch is just 0.5M params (0.6% of the model)** but contributed **+11.4 pts** of accuracy, more than scaling the spatial backbone from 28M to 89M params (which only added +2.1 pts).

97.39%

ACCURACY (TTA)

97.0%

PRECISION

~12 hr

TRAIN TIME • 25 EPOCHS

PER-FAKE-TYPE ACCURACY



0.9740

F1-SCORE (WEIGHTED)

97.0%

RECALL

~68 ms

INFERENCE / SAMPLE (single-pass)

Pure unsupervised anomaly detection collapsed to **near-chance**: AUC 0.54.

CONVOLUTIONAL AUTOENCODER

Train on real only • Score on residual

128×128 input • 16 base channels • 128-dim latent • **no skip connections** (so the decoder cannot just copy fakes through). Composite loss = L1 + SSIM + edge.

ANOMALY SCORE

P95 pixel error • SSIM error • latent distance from real centroid → z-scored, summed, thresholded by Youden's J on validation.

WHY IT UNDERPERFORMED

Modern deepfakes preserve the **global face manifold** ; reconstruction MSE is too coarse to expose sub-pixel manipulation. Honest negative result.

50.65%

ACCURACY

~2 hr

TRAIN TIME • 25 EPOCHS

45.27%

PRECISION

23.66%

RECALL

0.3108

F1-SCORE

~0.53 ms

INFERENCE / SAMPLE (single-pass)

CONFUSION MATRIX (TEST SET, 7,443 SAMPLES)

	PRED REAL	PRED FAKE
REAL	2,942 TN • 39.5%	1,001 FP • 13.4%
FAKE	2,672 FN • 35.9%	828 TP • 11.1%

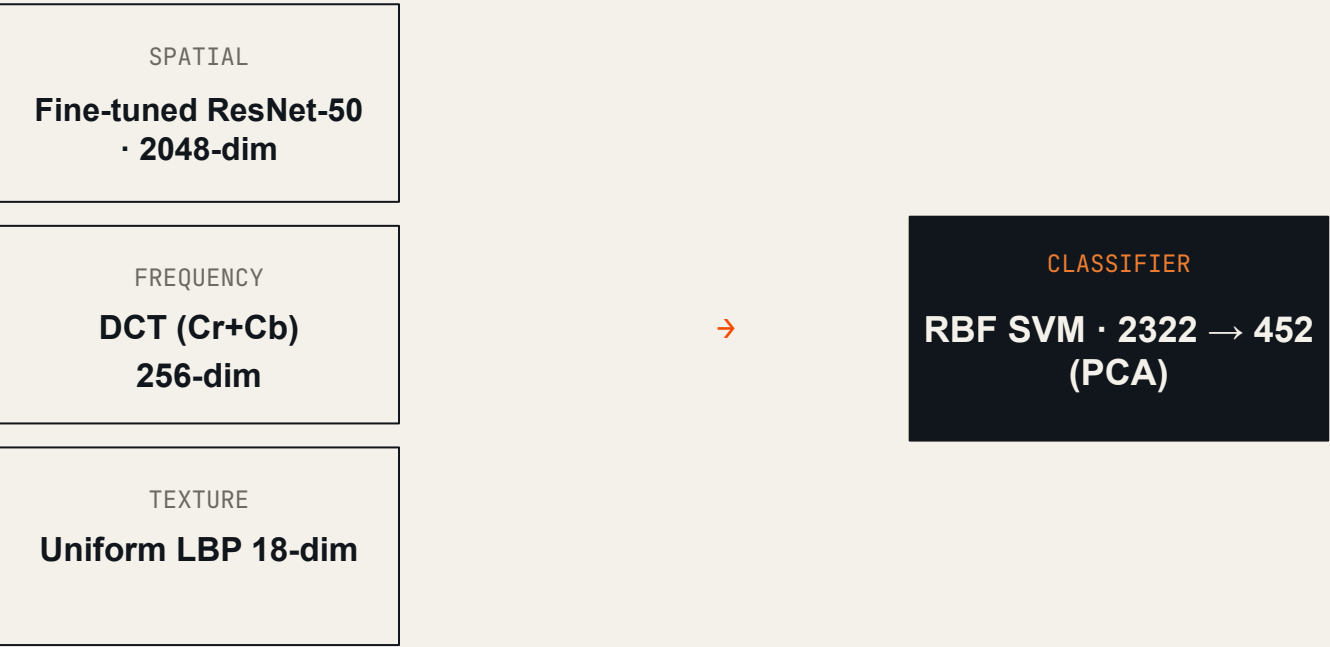
Every fake type lands within 7 pts of chance – the autoencoder learns "face" but not "fake".

Classical hybrid SVM hit **93.62%** — within **~1.37 pt** of the SOA, on a fraction of the compute.

CLASSICAL ML • FEATURE FUSION

Fine-tuned ResNet-50 + DCT + LBP → RBF SVM

Three parallel feature blocks fused into a 2,322-dim vector, scaled per-block, reduced to 452 components via PCA-95%, fed to an RBF SVM (C=0.2, $\gamma=1e-4$). GroupKFold by video ID prevents frame leakage. ResNet-50 layer4 was fine-tuned on the deepfake task before extraction.



SURPRISING FIND

Fine-tuning ResNet-50's layer4 alone added **+25 pts** over the frozen-CNN baseline (~69% → 93.62%) and collapsed the train/test gap from ~24 → ~5.5 pts. The bottleneck was feature quality, not the SVM.

93.62%

ACCURACY

94%

PRECISION (WEIGHTED)

94%

F1-SCORE (WEIGHTED)

94%

RECALL (WEIGHTED)

CONFUSION MATRIX (TEST SET, 40,527 SAMPLES)

	PRED FAKE	PRED REAL
FAKE	29,220 TP • 95.4%	1,407 FN • 4.6%
REAL	1,179 FP • 11.9%	8,721 TN • 88.1%

Train/test gap closed from ~24 → ~5.5 pts on the expanded master split.

MesoInception-4 — Targeting Mesoscopic Artifacts via Lightweight Inception.

LIGHTWEIGHT CNN • MESOSCOPIC

Multi-scale Inception modules at 384²

Two Inception modules (parallel 1×1, 3×3, 5×5 filters) feed two 5×5 conv + max-pool blocks. Tiny dense head: **16 units** → **dropout 0.5** → **LeakyReLU** → **sigmoid**. Optimized via **Adam**, **BCE**, **mixed precision**.

WHY MESOSCOPIC

Macroscopic geometry now looks perfect; **Microscopic** noise dies under JPEG. Mid-scale blending boundaries and texture gradients **survive compression**.

SURPRISING FIND

98.2% Accuracy on Celeb-synthesis (Near-perfect detection) and **+27% Boost** in NeuralTextures detection compared to Mid-term results.

83.35%

ACCURACY

0.88

ROC-AUC

84.39%

PRECISION (WEIGHTED)

83.35%

RECALL (WEIGHTED)

83.31%

F1-SCORE (WEIGHTED)

~34 ms

INFERENCE / SAMPLE

PER-CLASS ACCURACY — EASY VS. HARD



CLIP ViT-L/14 with GenD + D³ + PatchFFT — strongest generalization across fake types.

FOUNDATION MODEL • TRI-BRANCH HYBRID

Approach: Tri-Branch Foundation Model (CLIP ViT-L/14 Hybrid)

Key Architecture Feature

Combines semantic, structural (D³), and spectral (PatchFFT) analysis into a single fused head to capture multiple dimensions of video forgery.

Key Observation

While highly accurate at catching synthetic fakes (99.8% on Celeb-synthesis), the model struggles to confidently clear genuine, uncompressed footage (85.5% on real FF++)—a vulnerability unique to this architecture.

SURPRISING FIND

While highly accurate at catching synthetic fakes (99.8% on Celeb-synthesis), the model struggles to confidently clear genuine, uncompressed footage (85.5% on real FF++, 90 real youtube, and 94.5 celeb real) a vulnerability unique to this architecture.

94.99%

ACCURACY

89.17%

F1-SCORE

89.95%

PRECISION

89.38%

RECALL

~4 hr

TRAIN TIME • 25 EPOCHS

12.04ms

INFERENCE / SAMPLE

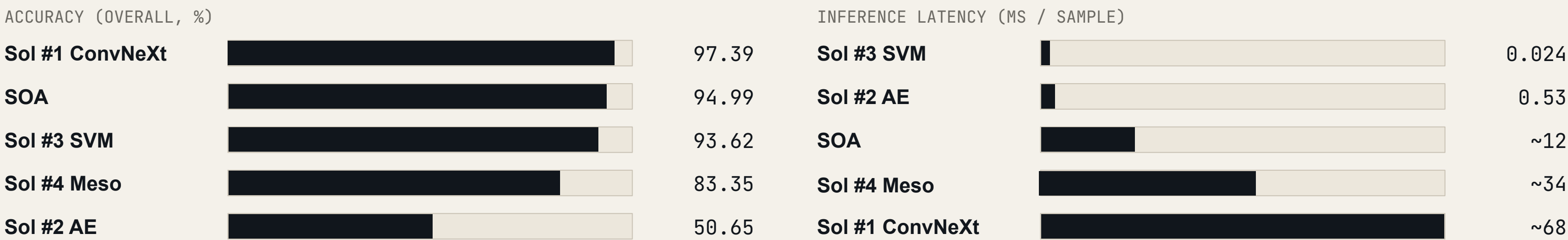
CONFUSION MATRIX (TEST SET, 42,927 SAMPLES)

	PRED FAKE	PRED REAL
FAKE	31,928 TP • 96.7%	1,099 FN • 3.3%
REAL	1,051 FP • 18.6%	8,849 TN • 89.4%

09 • OVERALL PERFORMANCE COMPARISON

Five solutions side-by-side. ConvNeXt V2 wins accuracy; the SVM wins value.

SOLUTION	ACCURACY	PRECISION	RECALL	F1	TRAIN TIME	INFERENCE	PARAMS
Sol #1 • ConvNeXt V2 + FFT	97.39%	97.0%	97.0%	0.974	~12 hr	~68 ms	88.2 M
Sol #2 • Autoencoder Anomaly	50.65%	45.27%	23.66%	0.311	~2 hr	0.53 ms	0.4 M
Sol #3 • Hybrid SVM Fusion	93.62%	93.8%	93.6%	0.935	~49 min	0.024 ms	25.6 M
Sol #4 • Mesoinception-4	83.35%	84.39%	83.35%	0.833	~3 hr	~34 ms	0.05 M
SOA • CLIP + GenD + D3 + FFT	94.99%	89.95%	89.38%	0.891	~4 hr	~12.04 ms	304 M



A 25.6M-param SVM matched a 304M-param foundation model — within 1.37 pts.

We expected the CLIP-based SOA (304M params, hyperspherical metric learning, patch-shuffle, FFT branch) to dominate Solution #3. Instead, the classical RBF SVM landed at **93.62%** versus **94.99%** — **a 1.37-pt gap on 12× fewer params and 14× less training time** .

VISUAL EVIDENCE — ACCURACY VS. COMPUTE

MODEL	ACC	PARAMS	TRAIN TIME
SOA · CLIP+GenD+D3+FFT	94.99%	304 M	~5hr
Sol #3 · SVM Fusion	93.62%	25.6 M	~48 min
Δ vs SOA	-1.37 pt	-92%	-93%

SVM training on CPU; ResNet50 feature extraction on a single T4.

SCIENTIFIC JUSTIFICATION — ML THEORY

Feature space already does the heavy lifting. A frozen ImageNet-pretrained ResNet50 already encodes enough discriminative texture and structure that the boundary between real and fake faces is approximately linear in the right kernel space — precisely what an RBF SVM is designed to exploit.

Bias–variance favours the simple model at our sample size. With only ~6K independent video groups after GroupKFold pooling, a 304M-param ViT operates in an over-parameterised regime relative to the data. The SVM with C=0.3 imposes strong margin regularisation, lowering variance with negligible bias cost.

Complementary handcrafted features beat scale alone. DCT + LBP added **+6.6 pts** over ResNet50 — orthogonal signals the deep features missed.

Every model — even the SOA — hits the same wall: NeuralTextures.

NeuralTextures is the **universal hardest class** . The ranking inverts everywhere else: macroscopic swaps (FaceSwap, FaceShifter, Celeb-synthesis) sit near 99%, but NT carves (**4–27 pts**) off every model — not just the weak ones.

PER-MODEL NT ACCURACY VS. EASIEST CLASS

MODEL	EASIEST CLASS	NT ACCURACY	Δ
Sol #1 ConvNeXt+FFT	99.8%	95.4%	-4.4
SOA CLIP hybrid	99.8%	91.3%	-8.5
Sol #3 SVM Fusion	~96%	~85%	-11
Sol #4 MesolInception	98.2%	70.7%	-27.5
Sol #2 Autoencoder	57.3%	54.9%	flat

NT modifies only mid-frequency texture — it leaves geometry, blending, and high-frequency spectra *untouched*.

SCIENTIFIC JUSTIFICATION — ML THEORY

Inductive biases miss the manipulation manifold. NeuralTextures replaces facial textures while *preserving* identity geometry, blending boundaries, and high-frequency spectra. Models keyed to those signals (PatchFFT, mesoscopic blending) have nothing to fire on.

Compression destroys the residual cue. H.264 quantisation in C23 collapses the mid-frequency band where NT artifacts live — what survives is below the noise floor of any single-branch detector.

Why ConvNeXt+FFT still cracks it (95.4%). The dual-branch design plus a *3× hard-mining multiplier* on NT during training forces the spatial branch to learn texture statistics the FFT branch can no longer see. Architecture diversity > model size.

Beyond accuracy: who would you actually deploy?

SOLUTION	INTERPRETABILITY	DEPLOYMENT COMPLEXITY	MAINTENANCE EFFORT	HARDWARE FLOOR	BEST REAL-WORLD FIT
Sol #1 ConvNeXt+FFT	Low	Medium	Medium	GPU (8 GB+)	Newsroom batch verification
Sol #2 Autoencoder	High	Low	Low	CPU	OOD / drift monitoring only
Sol #3 SVM Fusion	High	Low	Low	CPU	Production / regulated KYC
Sol #4 MesolInception	Medium	Low	Medium	Mobile / edge	On-device pre-screen
SOA CLIP hybrid	Low	High	High	Server GPU (24 GB+)	Research / forensic specialists

WHY THIS MATTERS

A KYC analyst who has to **justify a rejection in court** can point to the exact DCT band and LBP histogram bin that fired in the SVM. A 304M-param ViT cannot offer the same audit trail without expensive post-hoc tooling.

THE DEPLOYABLE WINNER ISN'T THE MOST ACCURATE

Solution #3 trails Solution #1 by **3.8 pts** but iterates ~7× faster once features are cached, with interpretable feature blocks. For deployments outside a research lab, the trade is worth it.

Lessons learned.

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- 01 **Architecture diversity beats parameter count.** A dual-branch 88M-param ConvNeXt+FFT outperformed a 304M ViT hybrid by **3.65 pts** . When two complementary inductive biases (spatial + spectral) are forced to disagree, the residual error shrinks faster than scaling either branch alone.
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- 02 **Classical ML is not dead — it's a viable production baseline.** Fine tuned ResNet50 + DCT + LBP fed to an RBF SVM hit **93.62%** at **1/12 the params** and **1/38 the training time** of the SOA. For any pipeline where compute, latency, or auditability matters, start here before reaching for a foundation model.
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- 03 **Pure unsupervised anomaly detection fails on modern deepfakes.** Reconstruction-based scoring landed at AUC 0.54 — modern generators preserve the real-face manifold too well. Anomaly methods belong in **drift monitoring**, not as the primary classifier.
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- 04 **Data leakage will silently inflate accuracy.** Frame-level splits put the same video in both train and test; the model memorizes lighting, compression, and identity instead of learning manipulation artifacts. **Stratify splits by source video** before any training run.

For the next team picking up this problem.

FOR DEEPPFAKE DETECTION RESEARCHERS

- Start with a **classical-feature SVM baseline** — if a foundation model can't clear it by 5+ pts, the gain is illusory.
- Add a **frequency branch** on synthetic data — the highest-leverage architectural change in the project.
- Avoid pure reconstruction anomaly** — modern generators stay on-manifold.
- Treat **NeuralTextures as hard-mined** from epoch 1.

FOR COMPARATIVE MULTI-SOLUTION PROJECTS

- Lock the data pipeline before anyone trains.** A shared, video-stratified split saves weeks of arguments.
- Budget 3× the time** for cross-metric reconciliation.
- Force **orthogonal inductive biases** — don't let four people build CNNs.

15 • CONCLUSION

Five detectors, one shared 360K-image pipeline, one clear answer: **a 25.6M-param classical SVM** trails a 304M foundation model by 1.37 points — and the dual-branch ConvNeXt+FFT beats both by leaning on **two signals**, not more parameters.

BEST OVERALL SOLUTION

Sol #1 (ConvNeXt+FFT) for accuracy-first deployments · Sol #3 (SVM Fusion) for production / regulated settings . We do not recommend a single winner — we recommend matching the model to the deployment constraint.

UNIQUE CONTRIBUTION OF COMPARISON

A single-solution project would have shipped Sol #1 and called it done. The comparative approach surfaced the SVM's value, the autoencoder's failure, and the universal NT ceiling — three findings only visible side-by-side.

FUTURE DIRECTIONS

→ Late-fusion ensemble of Sol #1 + Sol #3 logits.

→ Cross-generator OOD evaluation on diffusion deepfakes.

→ Distillation: SOA → MesolInception for edge deployment.



THANK YOU.

QUICK REFERENCE – FINAL SCOREBOARD

SOLUTION	ACC	F1	INFERENCE	DEPLOY FIT
Sol #1 ConvNeXt+FFT	97.39%	0.974	68 ms	Newsroom
SOA CLIP hybrid	93.74%	0.929	12.04 ms	Forensic lab
Sol #3 SVM Fusion	93.62%	0.935	0.024 ms	Production / KYC
Sol #4 Mesolnception	83.35%	0.833	34 ms	Edge / mobile
Sol #2 Autoencoder	50.65%	0.311	0.53 ms	Drift monitor only

REPO & REPORT

FakeBusters · CCAI 323 · Spring 2026
Final report · Final code freeze May 3.